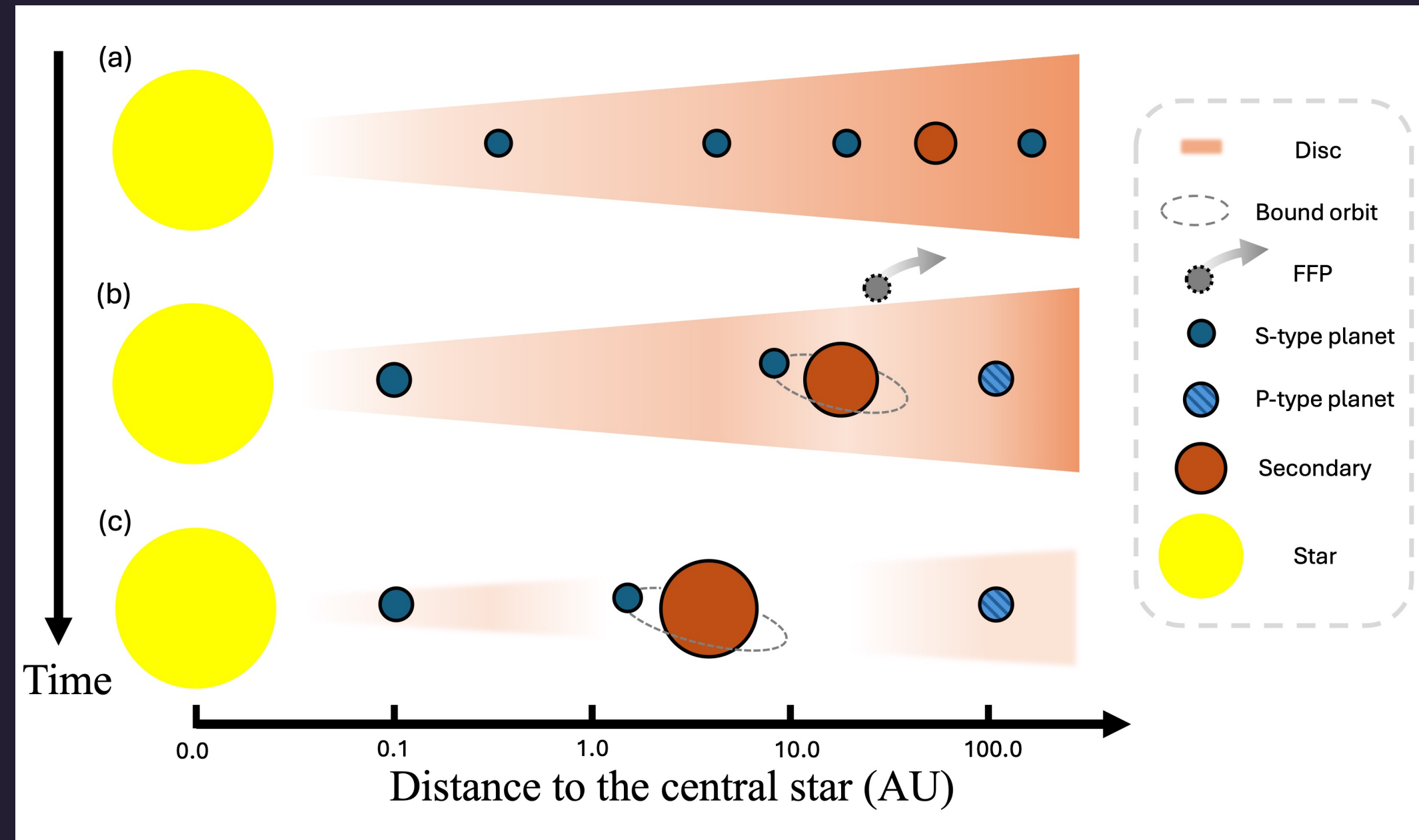




Disc Fragmentation: formation of close binaries, their planets, and free floating planet ejection

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Dozens of planets and brown dwarfs orbit one component of tight stellar binaries^[1], despite circumstellar discs in these systems being truncated to only a few au. This challenges classical planet formation models, which assume planets form after their host stars within stable discs. We propose instead that planet and binary formation are concurrent outcomes of gravitational fragmentation in massive circumstellar discs. In this scenario, rapid disc growth driven by cloud infall triggers fragmentation at tens of au, forming planetary-mass objects that migrate inward. Continued accretion leads a dominant “oligarch” fragment to undergo runaway growth and become the secondary star. Dynamical interactions then eject many lower-mass planets, producing free-floating planets (FFPs), while others survive if they migrate sufficiently close to the primary star before destabilisation.

1. A Fragmenting disc example

Simulations are performed with **FARGO-ADSG**^[2]. Successive clumps grow by disc fragmentation, with one eventually forming the secondary star. Sink particles are injected into clumps and spiral arms to represent collapsed solid cores.

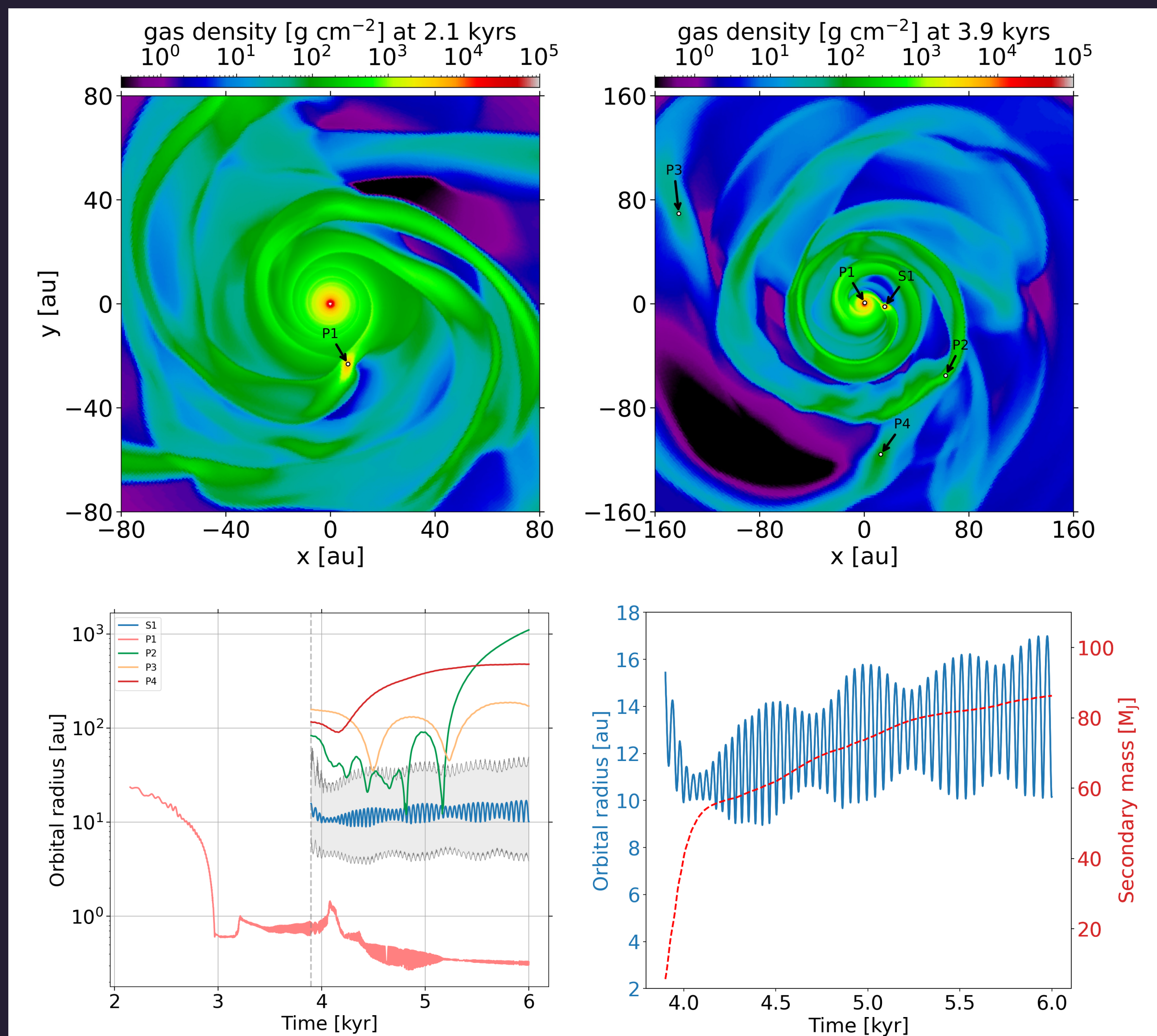


Fig. 1. **Top**: Disc surface density at the injection times of P1 (left) and secondary S1 with three additional cores (right). **Bottom**: Radial evolution of cores (left; shaded region indicates the Holman–Wiegert instability zone) and orbital radius and mass of the secondary (right).

2. A single embedded object

Examples of a single object evolving in a massive gravito-turbulent disc (non-fragmenting). One object is injected per disc at 60 au. For massive cases $M_p = 10, 12 M_J$, gas accretion is included.

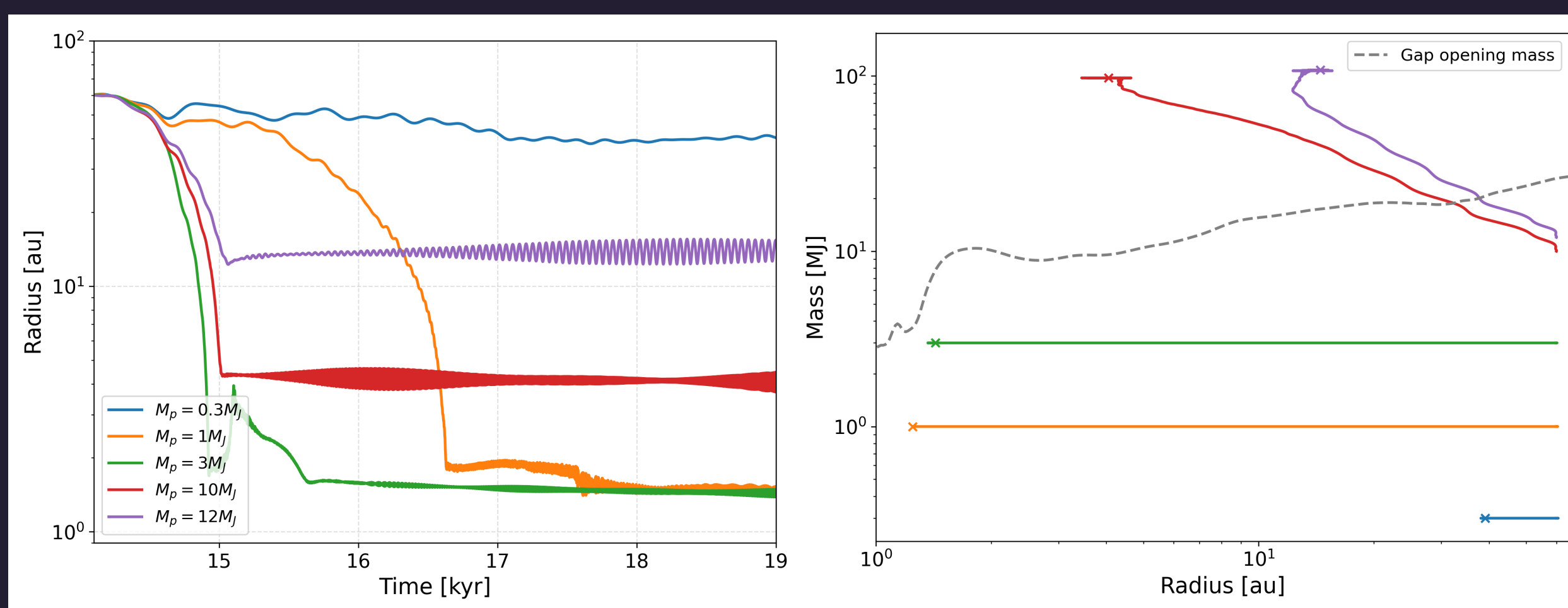


Fig. 2. **Left**: Evolution of the orbital radius for the secondary object with different masses. **Right**: Evolution of the object mass and radius in the mass-radius plane for the simulations shown on the left.

3. A grid of planet-secondary injection experiments

To model the secondary and planet in non-fragmenting disc, we inject:

- A $12 M_J$ secondary at 60 AU with an accretion radius of Hill radius.
- A planet placed at R_p with mass of M_p .

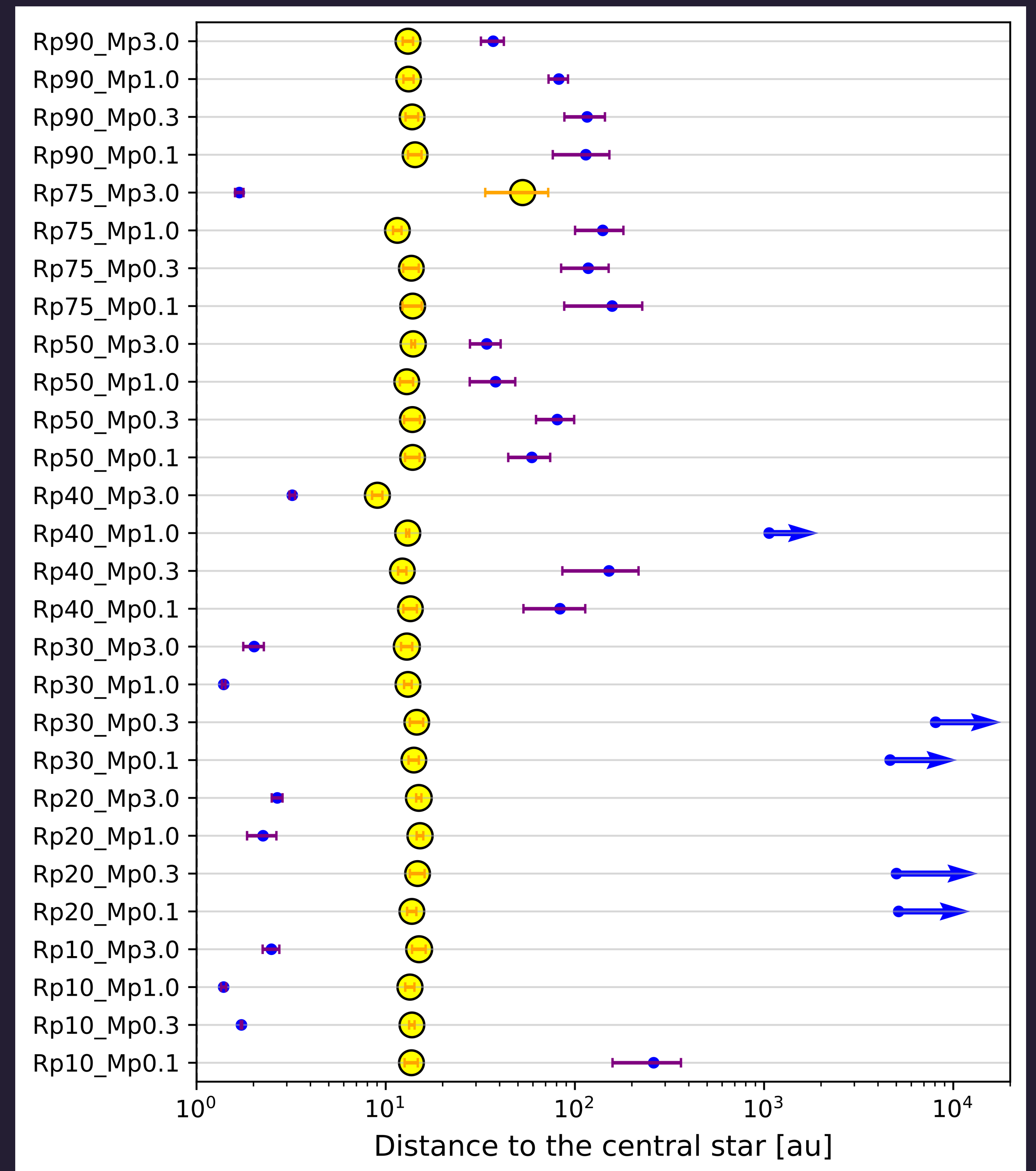


Fig 3. Census of simulated planet-hosting binaries. The y-axis lists simulations labeled as ‘**Rp**(initial planet orbital radius in au)_**Mp**(initial planet mass in M_J)’. Blue dots represent the planet with purple bars indicating eccentricity, while yellow dots represent the secondary companion with orange eccentricity bars. The size of the yellow dots is proportional to the final secondary mass. Arrows represent FFPs, with their length proportional to planet velocity.

Conclusions:

- **Planets may survive during the secondary growth process and remain bound to one of binary components**
Planets formed early with masses $\geq 1-3 M_J$ are preferentially retained, while lower-mass planets are typically ejected, which naturally explains why low-mass planets are more deficient in tight binaries.
- **FFPs should have a steeper mass function than bound planets within binaries.**
Due to the strong mass dependency of the ejection process, our scenario predicts that the mass function of the FFPs must be more bottom heavy than the one for planets in binaries.

References:

- [1] Thebault P., Bonanni D., 2025, A&A, 700, A106;
- [2] Baruteau, C., & Masset, F. 2008, ApJ, 678, 483